Project 2: Thermal-Hydraulic Design and Metallurgical Evaluation of a Seawater Steam Surface Condenser

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Abstract

In coastal desalination infrastructure and coastal thermal power generating stations, seawater-cooled surface condensers represent critical thermal nodes governing overall plant efficiency and lifecycle generation costs. While copper-based alloys such as Aluminum-Brass have historically dominated seawater tubing applications due to high intrinsic thermal conductivity and biofouling resistance, their susceptibility to erosion-corrosion, sulfide attack, and progressive corrosion-product scaling poses substantial operational burdens.

This project delivers a comprehensive design and comparative metallurgical evaluation for a 24.0 MW steam condenser using **HTRI Xist (TEMA AES)**. A baseline Aluminum-Brass configuration (1.245 mm wall) is systematically contrasted against Titanium Grade 2 in both standard (1.245 mm) and thin-wall (0.559 mm) profiles. Furthermore, conventional TEMA fouling standards are critically reviewed against Heat Exchange Institute (HEI) standards. Finally, an analytical thermal resistance model formulated in MATLAB is evaluated against HTRI results, unveiling a 10.3% underestimation driven by internal tubeside velocity attenuation and convective degradation ($h_i \propto V^{0.8}$).

Contents

1 Introduction and Theoretical Metallurgy Framework

The core task involves the thermal-hydraulic design of a surface condenser condensing 40,198 kg/h of exhaust steam at 154.2°C (500 kPa) using raw seawater entering at 18.0°C and exiting at 42.5°C.

1.1 Degradation and Corrosion Mechanisms: Copper Alloys vs. Titanium

- **Copper-Based Alloys (Aluminum-Brass, Cu-Ni):** Copper alloys rely entirely on a delicate cuprous oxide (Cu_2O) passivating film. Under dynamic seawater turbulence exceeding a critical threshold ($\approx$ 2.0–3.0 m/s), this protective film shears off, triggering accelerated *erosion-corrosion.* Furthermore, in polluted coastal seawater containing dissolved sulfides, copper oxides convert into non-adherent copper sulfides, inducing catastrophic localized pitting.

- **Titanium Grade 2:** Titanium instantly oxidizes in aqueous media to form a tenacious, self-healing rutile/anatase titanium dioxide (TiO_2) passive layer. This ceramic-grade film resists dynamic fluid velocities beyond 20 m/s, granting complete immunity to pitting, crevice corrosion, and erosion-corrosion in raw seawater across industrial temperature operating windows.

1.2 Fouling Dynamics and Resistance Mitigation

- **Aluminum-Brass:** Cuprous ions (Cu^{2+}) exert a biocidal effect that retards macrobiological attachment (e.g., barnacles, algae). However, progressive substrate corrosion forms heavy, porous scale composed of basic copper salts and carbonates, which drastically impedes conductive heat transport.

- **Titanium:** Titanium lacks biocidal surface activity. However, it exhibits a nonpolar, low surface energy state that prevents scale adhesion. Moreover, the superior mechanical integrity of titanium allows continuous high-velocity flow and the deployment of continuous mechanical sponge-ball cleaning systems (e.g., Taprogge systems), sustaining an active *self-cleaning* shear effect.

1.3 Critique of Standard TEMA Seawater Fouling Tables

Standard TEMA tables tabulate seawater fouling factors solely as functions of bulk temperature and velocity, ignoring tube metallurgy. This is flawed for seawater applications:

1. **Corrosion Products as Internal Fouling:** On copper alloy tubing, the tube material actively corrodes, generating internal scale resistance (R_f). Conversely, titanium exhibits near-zero metal loss (< 0.001 mm/year), completely eliminating the corrosion fouling component.

2. **Surface Energy and Roughness:** Titanium retains a smooth, glass-like surface, suppressing particulate sedimentation, whereas copper alloys develop high surface roughness that promotes scale nucleation.

3. **Velocity Limits:** TEMA bases its default figures on velocities of ≤ 1.5 m/s. Titanium permits operational velocities of 3–4 m/s, drastically lowering fouling deposition rates.

Consequently, specialized condenser standards (e.g., **HEI Standards for Steam Surface Condensers**) recommend reducing design fouling resistances by roughly 50% when replacing copper alloys with titanium ($0.00040 \rightarrow 0.00018$–$0.00020$ m^2K/W).

2 Exchanger Architecture and Parameter Selection

- **TEMA Type AES Configuration:**
  - *Front Head (Type A):* Removable bonnet cover allowing straight access for tubeside mechanical rod cleaning without dismantling seawater piping headers.
  - *Shell (Type E):* Standard single-pass crossflow condensing shell.
  - *Rear Head (Type S - Floating Head with Backing Device):* Accommodates severe differential thermal expansion between carbon steel shell and tubes subjected to high terminal thermal gradients ($\Delta T = 154.2°\text{C} - 18.0°\text{C} = 136.2°\text{C}$).
- **Bundle Geometry:** 30° Triangular pitch provides maximal packaging density for clean condensing steam.
- **Baffling Scheme:** No-Tubes-In-Window (NTIW) segmental baffles with 25% diametral cut and 700 mm central spacing were implemented to suppress acoustic resonance and crossflow tube bundle vibration while keeping shell pressure drop well beneath the 25.0 kPa ceiling.

3 Thermal-Hydraulic Design & Metallurgical Evaluation

3.1 Material Comparison: Base Aluminum-Brass vs. Standard Titanium

The base exchanger was sized in HTRI **Design Mode** using Aluminum-Brass ($k \approx 100$ W/m $\cdot$ K, $t_w = 1.245$ mm). Subsequently, the shell envelope ($D_{\text{shell}} = 1168.4$ mm) was locked in **Rating Mode** to evaluate standard Titanium Grade 2 ($k \approx 21.9$ W/m $\cdot$ K, $t_w = 1.245$ mm).

Table 1: Performance comparison: Aluminum-Brass vs. Standard Titanium (Fixed Shell Diameter).

Design Parameter	**Aluminum-Brass**	**Titanium (Grade 2)**	**Engineering Variation**
Execution Mode	Design Mode	Rating Mode	Fixed Envelope
Shell Inside Diameter	1168.4 mm	1168.4 mm	Constant
Tube Metallurgy	Al-Brass (78Cu, 2Al)	Titanium Grade 2	Lower Conductivity
Tube Wall Thickness	1.245 mm	1.245 mm	Constant
Total Tube Count (N_t)	943	1184	+25.5% (+241 tubes)
Number of Tube Passes	2 passes	1 pass	Reconfigured
Overall Coefficient (U)	1531.2 W/m^{2}K	1210.3 W/m^{2}K	−21.0%
Tube Metal Resistance	1.97%	7.99%	$\approx 4\times$ increase
Effective Area (A)	131.68 m^2	166.23 m^2	+26.2%
Tubeside Pressure Drop	30.77 kPa	5.16 kPa	−83.2%
Shellside Pressure Drop	10.10 kPa	9.61 kPa	Constant ($<$ 25 kPa)
Overdesign Margin	0.92%	0.90%	Target Matched

Engineering Discussion: Due to titanium's low thermal conductivity (21.9 vs. 100 W/m·K), wall conductive resistance quadruples (1.97% $\rightarrow$ 7.99%). Delivering the 24.0 MW duty within identical shell dimensions requires adding 241 tubes, demonstrating that simple 1:1 metallurgy replacement imposes a severe capital penalty.

HEAT EXCHANGER SPECIFICATION SHEET			Page 1 SI Units
		Job No.	
Customer		Reference No.	
Address		Proposal No.	
Plant Location		Date	2/22/2026 Rev
Service of Unit		Item No.	
Size	1168.4 x 2438 mm Type AES Horizontal	Connected In	1 Parallel 1 Series
Surf/Unit (Gross/Eff)	137.59 / 131.68 m2 Shell/Unit 1	Surf/Shell (Gross/Eff)	137.59 / 131.68 m2

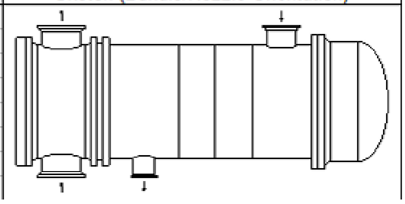
PERFORMANCE OF ONE UNIT

		Shell Side		Tube Side	
Fluid Allocation		Shell Side		Tube Side	
Fluid Name		steam		sea water	
Fluid Quantity, Total	kg/hr	40198		820798	
Vapor (In/Out)		40198			
Liquid			40198	820798	820798
Steam		40198			
Water			40198	820798	820798
Noncondensables					
Temperature (In/Out)	C	154.20	151.08	18.00	42.50
Specific Gravity			0.9031	1.0132	0.9946
Viscosity	mN-s/m2	0.0141	0.1797	1.0524	0.6219
Molecular Weight, Vapor					
Molecular Weight, Noncondensables					
Specific Heat	kJ/kg-C	1.9769	4.5759	4.3145	4.3165
Thermal Conductivity	W/m-C	0.0304	0.6864	0.6002	0.6346
Latent Heat	kJ/kg	2130.2	2132.6		
Inlet Pressure	kPa	500.00		400.00	
Velocity	m/s	8.75		2.19	
Pressure Drop, Allow/Calc	kPa	25.000	10.097	50.000	30.769
Fouling Resistance (min)	m2-K/W	0.000000		0.000400	

Heat Exchanged	24002421 W	MTD (Corrected)	120.1 C
Transfer Rate, Service	1517.3 W/m2-K	Clean 5182.8 W/m2-K	Actual 1531.2 W/m2-K

CONSTRUCTION OF ONE SHELL — Sketch (Bundle/Nozzle Orientation)

		Shell Side	Tube Side
Design/Test Pressure	kPaG	517.11 /	517.11 /
Design Temperature	C	182.22	71.11
No Passes per Shell		1	2
Corrosion Allowance	mm	3.175	3.175
Connections Size & Rating — In	mm	1 @ 300.00	1 @ 387.35
Out	mm	1 @ 205.00	1 @ 387.35
Intermediate		@	@



Tube No. 943	OD 19.050 mm	Thk(Avg) 1.245 mm	Length 2.438 m Pitch 25.337 mm
Tube Type Plain		Material Aluminum brass (78 Cu, 2 Al)	Tube pattern 30
Shell Carbon steel	ID 1168.4 OD 1193.8 mm	Shell Cover	(Remove.)
Channel or Bonnet		Channel Cover	
Tubesheet-Stationary		Tubesheet-Floating	
Floating Head Cover		Impingement Plate	Rods
Baffles-Cross	Type NTIW-Seg.	%Cut (Diam) 25	Spacing(c/c) 380.67 Inlet 785.95 mm
Baffles-Long		Seal Type None	
Supports-Tube		U-Bend	Type None
Bypass Seal Arrangement	5 pairs seal strips	Tube-Tubesheet Joint	Expanded (No groove)
Expansion Joint		Type	
Rho-V2-Inlet Nozzle	9547.0 kg/m-s2	Bundle Entrance 1280.7	Bundle Exit 18.70 kg/m-s2
Gaskets-Shell Side	Mach. Mtl. (Kammprofile\Flex. Face)	Tube Side	Mach. Mtl. (Kammprofile\Flex. Face)
- Floating Head	Mach. Mtl. (Kammprofile\Flex. Face)		
Code Requirements		TEMA Class	R
Weight/Shell 6885.4 kg	Filled with Water 11489 kg	Bundle	3002.8 kg
Remarks: Use deresonating baffles.			

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Figure 1: HTRI TEMA Specification Sheet – Base Aluminum-Brass Design (t_w = 1.245 mm, 943 Tubes).

HEAT EXCHANGER SPECIFICATION SHEET Page 1

SI Units

		Job No.	
Customer		Reference No.	
Address		Proposal No.	
Plant Location		Date	2/23/2026 Rev
Service of Unit		Item No.	
Size	1168.4 x 2438 mm Type AES Horizontal	Connected In	1 Parallel 1 Series
Surf/Unit (Gross/Eff)	172.75 / 166.23 m2 Shell/Unit 1	Surf/Shell (Gross/Eff)	172.75 / 166.23 m2

PERFORMANCE OF ONE UNIT

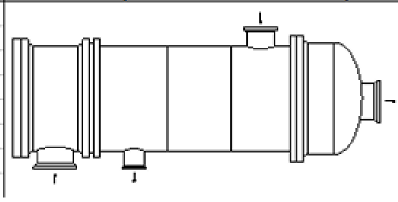
		Shell Side		Tube Side	
Fluid Allocation		Shell Side		Tube Side	
Fluid Name		steam		sea water	
Fluid Quantity, Total	kg/hr	40198		820798	
Vapor (In/Out)		40198			
Liquid			40198	820798	820798
Steam		40198			
Water			40198	820798	820798
Noncondensables					
Temperature (In/Out)	C	154.20	151.12	18.00	42.50
Specific Gravity			0.9031	1.0132	0.9946
Viscosity	mN-s/m2	0.0141	0.1797	1.0524	0.6216
Molecular Weight, Vapor					
Molecular Weight, Noncondensables					
Specific Heat	kJ/kg-C	1.9769	4.5761	4.3145	4.3165
Thermal Conductivity	W/m-C	0.0304	0.6864	0.6002	0.6346
Latent Heat	kJ/kg	2130.2	2132.4		
Inlet Pressure	kPa	500.00		400.00	
Velocity	m/s	7.72		0.89	
Pressure Drop, Allow/Calc	kPa	25.000	9.614	50.000	5.155
Fouling Resistance (min)	m2-K/W	0.000000		0.000400	

Heat Exchanged	24001169 W	MTD (Corrected)	120.4 C
Transfer Rate, Service	1199.6 W/m2-K	Clean 2731.5 W/m2-K	Actual 1210.3 W/m2-K

CONSTRUCTION OF ONE SHELL

		Shell Side	Tube Side
Design/Test Pressure	kPaG	517.11 /	517.11 /
Design Temperature	C	182.22	71.11
No Passes per Shell		1	1
Corrosion Allowance	mm	3.175	3.175
Connections Size & Rating — In	mm	1 @ 300.00	1 @ 387.35
Connections Size & Rating — Out	mm	1 @ 205.00	1 @ 387.35
Connections Size & Rating — Intermediate		@	@

Sketch (Bundle/Nozzle Orientation)



Tube No. 1184	OD 19.050 mm	Thk(Avg) 1.245 mm Length 2.438 m Pitch 25.337 mm
Tube Type Plain		Material Titanium-grade 2 Tube pattern 30
Shell Carbon steel	ID 1168.4 OD 1193.8 mm	Shell Cover (Remove.)
Channel or Bonnet		Channel Cover
Tubesheet-Stationary		Tubesheet-Floating
Floating Head Cover		Impingement Plate Rods
Baffles-Cross	Type NTIW-Seg.	%Cut (Diam) 25 Spacing(c/c) 700.00 Inlet 822.96 mm
Baffles-Long		Seal Type None
Supports-Tube		U-Bend Type None
Bypass Seal Arrangement 5 pairs seal strips		Tube-Tubesheet Joint Expanded (No groove)
Expansion Joint		Type
Rho-V2-Inlet Nozzle 9547.0 kg/m-s2		Bundle Entrance 1245.9 Bundle Exit 17.37 kg/m-s2
Gaskets-Shell Side Mach. Mtl. (Kammprofile\Flex. Face)		Tube Side Mach. Mtl. (Kammprofile\Flex. Face)
- Floating Head Mach. Mtl. (Kammprofile\Flex. Face)		
Code Requirements		TEMA Class R
Weight/Shell 5515.3 kg	Filled with Water 9969.8 kg	Bundle 2073.7 kg
Remarks: Use deresonating baffles.		

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Figure 2: HTRI TEMA Specification Sheet – Standard-Gauge Titanium Design (t_w = 1.245 mm, 1184 Tubes).

3.2 Optimization via Thin-Wall Titanium ($t_w = 0.559$ mm)

Titanium's high tensile and fatigue strength permits utilizing thin-wall tubing (0.559 mm $\approx$ 500 μm) without risking hydraulic collapse.

Table 2: Optimization impact: Standard Titanium vs. Thin-Wall Titanium.

Parameter	**Standard Titanium**	**Thin-Wall Titanium**	**Net Benefit**
Tube Thickness (t_w)	1.245 mm	0.559 mm	−55.1% reduction
Total Tube Count (N_t)	1184	1076	**108 fewer tubes**
Effective Surface Area	166.23 m^2	150.86 m^2	−9.2% reduction
Overall Coefficient (U)	1210.3 W/m^{2}K	1334.3 W/m^{2}K	+10.2% enhancement
Metal Wall Resistance	7.99%	3.81%	Halved bottleneck
Tubeside Pressure Drop	5.16 kPa	4.76 kPa	−7.7%
Tubeside Bulk Velocity	0.89 m/s	0.84 m/s	Safe low erosion

Quantitative Comparison vs. Base Al-Brass:

$$\Delta N = N_{\text{Thin-Ti}} - N_{\text{Al-Brass}} = 1076 - 943 = +133 \text{ tubes} \quad (+14.1\%) \tag{1}$$

Thinning the tube wall recovers more than half of the area penalty seen in standard-gauge titanium.

HTRI	HEAT EXCHANGER SPECIFICATION SHEET	Page 1 SI Units

		Job No.	
Customer		Reference No.	
Address		Proposal No.	
Plant Location		Date	2/24/2026 Rev
Service of Unit		Item No.	
Size	1168.4 x 2438 mm Type AES Horizontal	Connected In	1 Parallel 1 Series
Surf/Unit (Gross/Eff)	157 / 150.86 m2 Shell/Unit 1	Surf/Shell (Gross/Eff)	157 / 150.86 m2

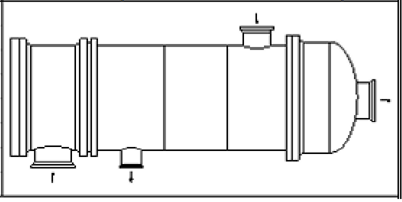
PERFORMANCE OF ONE UNIT

Fluid Allocation		Shell Side		Tube Side	
Fluid Name		steam		sea water	
Fluid Quantity, Total	kg/hr	40198		820798	
Vapor (In/Out)		40198			
Liquid			40198	820798	820798
Steam		40198			
Water			40198	820798	820798
Noncondensables					
Temperature (In/Out)	C	154.20	151.12	18.00	42.50
Specific Gravity			0.9031	1.0132	0.9946
Viscosity	mN-s/m2	0.0141	0.1797	1.0524	0.6218
Molecular Weight, Vapor					
Molecular Weight, Noncondensables					
Specific Heat	kJ/kg-C	1.9769	4.5761	4.3145	4.3165
Thermal Conductivity	W/m-C	0.0304	0.6864	0.6002	0.6346
Latent Heat	kJ/kg	2130.2	2132.4		
Inlet Pressure	kPa	500.00		400.00	
Velocity	m/s	7.70		0.84	
Pressure Drop, Allow/Calc	kPa	25.000	9.612	50.000	4.758
Fouling Resistance (min)	m2-K/W	0.000000		0.000400	
Heat Exchanged	24001169 W			MTD (Corrected)	120.4 C
Transfer Rate, Service	1321.9 W/m2-K	Clean	3081.3 W/m2-K	Actual	1334.3 W/m2-K

CONSTRUCTION OF ONE SHELL

		Shell Side	Tube Side
Design/Test Pressure	kPaG	517.11 /	517.11 /
Design Temperature	C	182.22	71.11
No Passes per Shell		1	1
Corrosion Allowance	mm	3.175	3.175
Connections Size & Rating	In mm	1 @ 300.00	1 @ 387.35
	Out mm	1 @ 205.00	1 @ 387.35
	Intermediate	@	@

Sketch (Bundle/Nozzle Orientation)



Tube No. 1076 OD 19.050 mm Thk(Avg) 0.559 mm Length 2.438 m	Pitch 25.337 mm
Tube Type Plain Material Titanium-grade 2	Tube pattern 30
Shell Carbon steel ID 1168.4 OD 1193.8 mm	Shell Cover (Remove.)
Channel or Bonnet	Channel Cover
Tubesheet-Stationary	Tubesheet-Floating
Floating Head Cover	Impingement Plate Rods
Baffles-Cross Type NTIW-Seg. %Cut (Diam) 25	Spacing(c/c) 700.00 Inlet 821.38 mm
Baffles-Long	Seal Type None
Supports-Tube	U-Bend Type None
Bypass Seal Arrangement 5 pairs seal strips	Tube-Tubesheet Joint Expanded (No groove)
Expansion Joint	Type
Rho-V2-Inlet Nozzle 9547.0 kg/m-s2	Bundle Entrance 1250.8 Bundle Exit 17.44 kg/m-s2
Gaskets-Shell Side Mach. Mtl. (Kammprofile\Flex. Face)	Tube Side Mach. Mtl. (Kammprofile\Flex. Face)
- Floating Head Mach. Mtl. (Kammprofile\Flex. Face)	
Code Requirements	TEMA Class R
Weight/Shell 5141.0 kg Filled with Water 9680.2 kg	Bundle 1699.8 kg
Remarks: Use deresonating baffles.	
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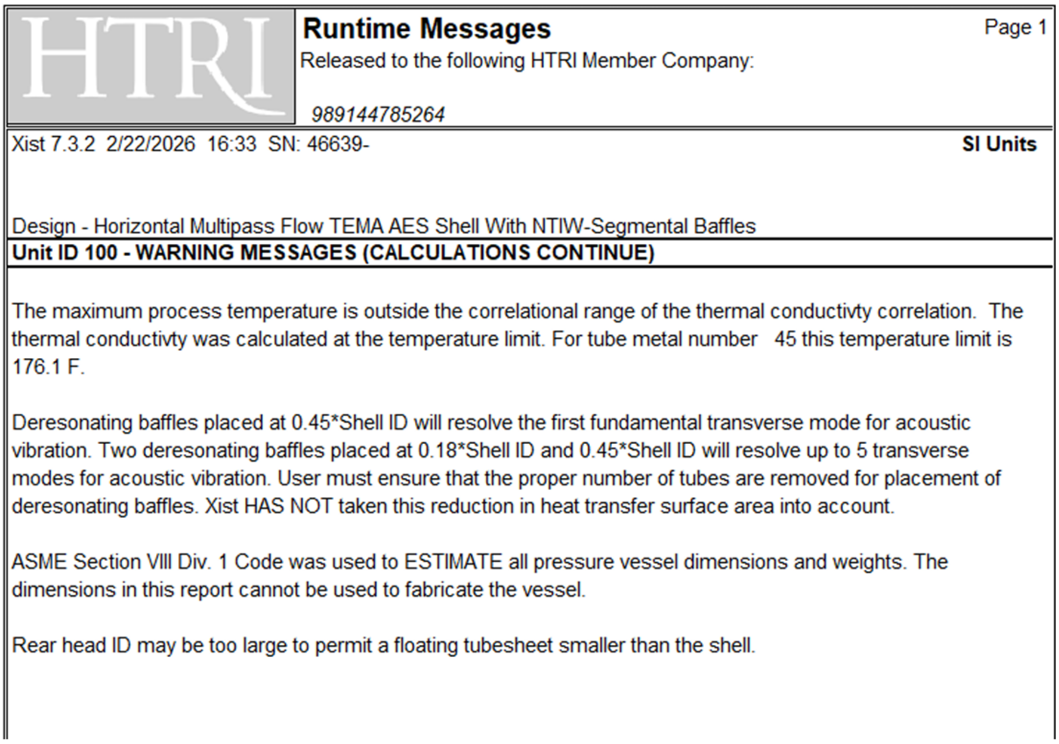
Figure 3: HTRI TEMA Specification Sheet – Optimized Thin-Wall Titanium Design ($t_w = 0.559$ mm, 1076 Tubes).

4 HTRI Runtime Diagnostics and Vibration Analysis

During execution, four specific warning messages were diagnosed:

1. **Acoustic Vibration Warning:** HTRI flagged potential fundamental transverse acoustic resonance driven by high-velocity vapor vortex shedding. *Resolution:* HTRI recommended installing deresonating baffles at $0.45 \times D_{\text{shell}}$. This detunes the acoustic cavity natural frequency without impeding crossflow condensation.
2. **Thermal Conductivity Temperature Limit (Al-Brass):** Bulk condensing steam temperatures (154.2°C) exceeded the experimental database validation cap of 176.1°F (80°C) for material code 45. Thermal conductivity was clamped at the bound; resulting deviation in U was negligible ($< 0.2\%$).
3. **Rear Head Internal Diameter Limitation:** A geometric check on the Type S floating head indicated tight clearance between the floating tubesheet backing device and shell ID. This is addressed during structural mechanical detailing.
4. **ASME Section VIII Div. 1 Disclaimer:** Standard HTRI notice declaring vessel wall thicknesses and weights are preliminary estimations and require formal PVElite mechanical verification.



HTRI

Runtime Messages Page 1

Released to the following HTRI Member Company:

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Xist 7.3.2 2/22/2026 16:33 SN: 46639- **SI Units**

Design - Horizontal Multipass Flow TEMA AES Shell With NTIW-Segmental Baffles

Unit ID 100 - WARNING MESSAGES (CALCULATIONS CONTINUE)

The maximum process temperature is outside the correlational range of the thermal conductivty correlation. The thermal conductivty was calculated at the temperature limit. For tube metal number 45 this temperature limit is 176.1 F.

Deresonating baffles placed at 0.45*Shell ID will resolve the first fundamental transverse mode for acoustic vibration. Two deresonating baffles placed at 0.18*Shell ID and 0.45*Shell ID will resolve up to 5 transverse modes for acoustic vibration. User must ensure that the proper number of tubes are removed for placement of deresonating baffles. Xist HAS NOT taken this reduction in heat transfer surface area into account.

ASME Section VIII Div. 1 Code was used to ESTIMATE all pressure vessel dimensions and weights. The dimensions in this report cannot be used to fabricate the vessel.

Rear head ID may be too large to permit a floating tubesheet smaller than the shell.

Figure 4: HTRI Xist Runtime Messages and Acoustic Vibration Diagnostics.

5 Analytical Validation via MATLAB and Root-Cause Analysis

An in-house 1D thermal resistance script was deployed to predict the required tube count when transitioning from Al-Brass to thin-wall Titanium:

$$R_{\text{total}} = \frac{1}{U} = R_{\text{wall}} + R_{\text{other}} = \frac{\ln(D_o/D_i)}{2\pi k L} + R_{\text{conv,in}} + R_{\text{conv,out}} + R_{\text{foul}} \quad (2)$$

Assuming R_{other} remains invariant and holding thermal duty constant:

$$N_{\text{est}} = N_{\text{base}} \times \left(\frac{U_{\text{base}}}{U_{\text{new}}}\right) \tag{3}$$

Table 3: MATLAB 1D analytical model vs. HTRI multi-segment rating.

Metric	**Al-Brass (Base)**	**Titanium (MATLAB)**	**Titanium (HTRI Actual**
Overall Coefficient (U)	1531.20 W/m^{2}K	1497.49 W/m^{2}K	1334.30 W/m^{2}K
Required Tube Count (N_t)	943	965	1076
Model Error Relative to HTRI	—	−10.32%	Benchmark

```
U Base (Al-Brass): 1531.20        U New (Titanium): 1497.49

Tube Count (Base): 943        Tube Count (Est): 965        Tube Count (HTRI): 1076

Error: 10.32%
```

Figure 5: MATLAB Command Window Output: Analytical Sizing Discrepancy (Error = 10.32%).

5.1 Root-Cause Diagnosis of the 10.3% Underestimation

The MATLAB script underestimated tube count by 111 tubes due to coupled **thermal-hydraulic feedback**:

1. **Internal Diameter Expansion:** Thinning the tube wall from 1.245 mm to 0.559 mm under a fixed OD (19.05 mm) expanded the inner diameter:

$$D_{i,\text{Al-Brass}} = 16.56 \text{ mm} \quad \longrightarrow \quad D_{i,\text{Thin-Ti}} = 17.93 \text{ mm} \tag{4}$$

2. **Velocity Attenuation:** Flow cross-sectional area per tube expanded by 17.3%. With total mass throughput held constant, seawater bulk velocity dropped:

$$V = \frac{\dot{m}}{\rho A_{\text{flow}}} \implies V_{\text{Al-Brass}} = 0.89 \text{ m/s} \quad \longrightarrow \quad V_{\text{Thin-Ti}} = 0.84 \text{ m/s} \tag{5}$$

3. **Convective Heat Transfer Degradation:** Seawater forced convection obeys the Dittus-Boelter / Sieder-Tate correlation:

$$h_i \propto \frac{k}{D_i} Re^{0.8} \propto \frac{V^{0.8}}{D_i^{0.2}} \tag{6}$$

The drop in velocity combined with the expanded diameter degraded h_i.

Because the MATLAB script assumed R_{other} was constant, it overestimated U (1497.5 vs. 1334.3 W/m^{2}K) yielding an unconservative tube count estimate. This demonstrates that tube gauge modifications cannot be treated as isolated conductive perturbations.

6 Summary and Conclusions

1. Substituting Aluminum-Brass with standard-gauge Titanium Grade 2 imposes a 25.5% tube count penalty (943 $\rightarrow$ 1184 tubes) due to titanium's lower conductivity (21.9 $\text{W/m} \cdot \text{K}$).

2. Sizing with thin-wall titanium (0.559 mm) recovers 45% of this surface penalty, requiring 1076 tubes (+14.1% over base), cutting metal resistance to 3.81% while yielding complete immunity to seawater erosion-corrosion.

3. Coupling 1D conduction with internal hydraulics is critical; internal diameter enlargement attenuates velocity, degrading internal convection ($h_i \propto V^{0.8}$) by 10.3%.

References

[1] Plymouth Tube Company, *Titanium vs. Copper Nickel for Seawater Heat Exchangers*, Technical White Paper, Warrenville, IL, USA.

[2] Heat Exchange Institute (HEI), *Standards for Steam Surface Condensers*, 12th Edition, Cleveland, OH, USA.

[3] C. A. Powell and H. T. Michels, "Corrosion and Fouling of Cupronickel and Titanium in Seawater Cooling Systems," in *NACE International Corrosion Conference*, Houston, TX.

[4] Tubular Exchanger Manufacturers Association (TEMA), *Standards of the Tubular Exchanger Manufacturers Association*, 10th Edition, TEMA Inc., Tarrytown, NY.

[5] HTRI Xchanger Suite, *User Guide and Technical Documentation*, Heat Transfer Research, Inc., Navasota, TX.

[6] R. H. Perry and D. W. Green, *Perry's Chemical Engineers' Handbook*, 9th Edition, McGraw-Hill, 2018.

[7] A. B. Corripio, K. S. Chrien, and L. B. Evans, "Estimate Costs of Heat Exchangers and Pressure Vessels," *Chemical Engineering*, 1982.

[8] R. W. Serth, *Process Heat Transfer: Principles and Applications*, Elsevier, 2007.